

**ERA5-Augmented Ensemble Learning for Multi-Horizon Tropical Cyclone
Track Prediction over the Bay of Bengal: A Leakage-Aware Benchmark**

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6 ABSTRACT: Accurate short- to medium-range cyclone track forecasting is a foundational input to
7 early warning over the Bay of Bengal, a basin that combines complex monsoon-season steering with
8 some of the highest coastal population exposure in the world. We present a rigorously validated
9 benchmark of eight track-prediction systems, a proposed Subset-Expert Context-Aware Ensemble
10 (SECE v2 Phase 3) and seven baselines spanning tree and hybrid families, evaluated at 3, 12, 24,
11 and 48 h lead times on North Indian Ocean IBTrACS storms with genesis in 1940–2024. After
12 quality control the modelling table contains 852 storms and 27,728 three-hourly forecast origins, of
13 which 583 storms and 15,605 origins support multi-horizon supervision. Forty-nine kinematic and
14 derived predictors are augmented with 14 ERA5 reanalysis fields collocated at the storm centre,
15 comprising winds at 850, 500, and 200 hPa, mean sea level pressure, sea surface temperature, a
16 derived deep-layer steering wind, and vertical wind shear. Critically, we identify and correct a
17 subtle form of test-set leakage present in an earlier iteration of our own development, in which
18 architectural decisions were guided by repeated inspection of test-set performance; we replace it
19 with a two-tier protocol in which nine development seeds carry all design decisions and nine disjoint
20 held-out seeds are evaluated exactly once. On the held-out confirm, pooled over 442 distinct test
21 storms, SECE attains median great-circle errors of 4.895, 35.538, 101.186, and 246.331 km. Paired
22 Wilcoxon signed-rank tests with Bonferroni correction across seven comparisons show that SECE
23 is significantly better than every baseline at 3 and 12 h ($p < 0.0071$), statistically indistinguishable
24 from the strongest stacks at 24 and 48 h, and never significantly worse than any baseline at any
25 horizon. Controlled ablations show that ERA5 features deliver genuine but horizon-dependent
26 value, concentrated at 48 h on a per-storm basis, while 5° and 3° spatial patch statistics were tested
27 and not adopted.

28 SIGNIFICANCE STATEMENT: This study provides a leakage-corrected, statistically tested
29 benchmark of tree-based tropical cyclone track models for the Bay of Bengal at 3 to 48 h lead
30 times. It shows where a routed ensemble is genuinely better than strong baselines, where it is not,
31 and where ERA5 environmental fields do and do not reduce track error, so agencies without a full
32 numerical weather prediction suite can interpret machine-learning guidance honestly.

33 1. Introduction

34 The Bay of Bengal has produced a disproportionate share of the deadliest tropical cyclones in
35 recorded history. Its shallow bathymetry, funnel geometry, and low lying deltaic margin convert
36 moderate storms into severe surge events, and the population density along that margin means that
37 forecast track error translates directly into evacuation scope and timing (World Bank 2007; Paul
38 2013; Dube et al. 2009). Reliable guidance at lead times of a few hours to two days is therefore not
39 an academic refinement but an operational requirement.

40 Numerical weather prediction remains the reference standard for that guidance, and dynamical-
41 model consensus is a mature part of operational practice (Goerss 2000, 2004). Its computational
42 and assimilation demands, however, are frequently beyond the reach of national meteorological
43 agencies in developing countries. Data-driven approaches that learn directly from historical best-
44 track archives offer a lightweight complement: the archives are public and well-curated (Knapp et al.
45 2010; Gahtan et al. 2024), and global reanalysis now supplies a physically consistent reconstruction
46 of atmospheric state at every historical storm position (Hersbach et al. 2023), so a learned model
47 can condition on environmental context without running a forecast cycle (Giffard-Roisin et al.
48 2020; Kumar et al. 2024).

49 Published approaches span a wide methodological range. Climatological-persistence analogues
50 of the CLIPER family remain standard references (Neumann 1972; Aberson 1998); recurrent and
51 convolutional sequence models have been applied extensively to track prediction (Lian et al. 2020;
52 Alemany et al. 2019; Rahman et al. 2025); and transformer architectures show promise at larger
53 scales (Qu et al. 2025). At the sample sizes available for a single basin, where storm counts are in
54 the hundreds rather than the millions, tree ensembles over engineered features nonetheless remain
55 difficult to displace (Grinsztajn et al. 2022; Ganaie et al. 2022).

56 Three substantive gaps persist specifically for the North Indian Ocean. First, architectural
57 benchmarking is narrow: most studies compare a small number of model families, so claims of
58 relative superiority rest on thin comparisons (Kar and Banerjee 2021; Hossain et al. 2021). Second,
59 evaluation is usually confined to a single lead time, which leaves horizon-dependent behaviour
60 undocumented even though the operational value of a forecast changes sharply with lead (Kumar et
61 al. 2024). Third, the incremental contribution of atmospheric reanalysis beyond kinematic features
62 has not, to our knowledge, been isolated for this basin by controlled single-variable ablation.

63 A fourth gap is methodological, and it motivates a central contribution of this paper. Published
64 ensemble benchmarks rarely disclose, or guard against, the leakage introduced by iterative human-
65 guided architecture search. When a designer repeatedly inspects held-out performance and adds
66 components that close the gaps observed there, the architecture has been fitted to the test set
67 even though no model parameter ever saw a test label, and the resulting held-out estimate is
68 optimistic (Cawley and Talbot 2010). We encountered exactly this failure mode in our own
69 development, and we document it here together with the diagnostic that exposed it and the protocol
70 that replaced it.

71 *a. Contributions*

- 72 • An eight-system benchmark evaluated at four forecast horizons over the Bay of Bengal and
73 wider North Indian Ocean, under storm-wise splits, with bootstrap confidence intervals and
74 paired significance testing pooled over nine held-out seeds and 442 distinct test storms.
- 75 • SECE v2 Phase 3, an ensemble that combines physics-informed feature subsets, heterogeneous
76 tree learners, and a dedicated ERA5-derived atmospheric expert, fused through horizon-
77 specific non-negative least squares candidates selected by a validation-only router.
- 78 • A controlled, single-variable-at-a-time ablation isolating the effect of point ERA5 features
79 against track-only kinematics, of a dedicated atmospheric expert against dispersing ERA5 into
80 existing subsets, and of spatial patch features against single-point extraction.
- 81 • Full statistical validation rather than point-estimate comparison, together with a transpar-
82 ent account of the leakage failure mode encountered during development and the corrected
83 protocol adopted in response.

2. Related Work

Classical statistical methods form the baseline layer of operational cyclone forecasting. Climatology-and-persistence models of the CLIPER family remain reference systems against which skill is measured (Neumann 1972; Aberson 1998), and statistical-dynamical methods have long been applied to Bay of Bengal storms alongside surge modelling (Dube et al. 2009). Dynamical-model consensus, in which several independent forecasts are averaged, is the operational expression of the observation that combining diverse predictors reduces track error (Goerss 2000, 2004); the ensembles studied here can be read as a learned analogue of that idea.

Machine learning for cyclone trajectories has largely followed the deep learning trajectory of neighbouring fields. Recurrent architectures have been applied to hurricane and typhoon tracks (Alemany et al. 2019; Rahman et al. 2025), convolutional-recurrent hybrids with multi-dimensional feature selection have been proposed for typhoon tracks (Lian et al. 2020), fused deep learning over aligned reanalysis fields has been demonstrated for track forecasting (Giffard-Roisin et al. 2020), and non-iterative spatiotemporal transformers have recently been applied to intensity (Qu et al. 2025). Reviews of ensemble deep learning (Ganaie et al. 2022) and of machine learning for cyclone track and intensity (Kumar et al. 2024) survey this space.

Against that trend, Grinsztajn et al. (2022) show that tree-based methods consistently outperform deep learning on medium-sized tabular data with engineered features, a regime that describes basin-scale cyclone track prediction closely: storm counts here are in the hundreds, and the predictive signal is concentrated in a few dozen hand-constructed kinematic and environmental quantities. Our results extend that observation one level further. Even within the tree-based ensemble family, additional architectural complexity does not improve performance uniformly as lead time grows; a simpler stacking approach becomes statistically competitive with a more elaborate context-aware ensemble at longer leads.

Basin-specific work on the North Indian Ocean has tended to address either intensity classification from best-track data (Kar and Banerjee 2021), single-storm case studies with dynamical models (Hossain et al. 2021), or neural regression for track at a single horizon (Krishnamurti et al. 2010). Geographically focused machine-learning studies elsewhere reinforce the value of basin-specific modelling (Pan 2024), and recent global benchmarking efforts underline how much evaluation protocol matters when systems are compared (Gomez et al. 2026). To our knowledge

no prior published benchmark evaluates this breadth of model families across four horizons on Bay of Bengal data under a leakage-controlled, statistically validated protocol.

The locked held-out confirm excludes deep sequence baselines for the same empirical and practical reasons we document in section e: in exploratory development under the uniform training budget, CNN-GRU and BLSTM repeatedly trailed the tree and hybrid leaders at 3–24 h, consistent with Grinsztajn et al. (2022), and were not re-run on nine held-out seeds with ERA5 and 48 h supervision.

3. Data and Experimental Design

a. Source archives and study domain

Storm trajectories come from the International Best Track Archive for Climate Stewardship, version 4, North Indian Ocean basin (Knapp et al. 2010; Gahtan et al. 2024). Atmospheric state comes from ERA5, the fifth-generation ECMWF reanalysis, which provides physically consistent gridded fields at sub-daily resolution continuously back to 1940 (Hersbach et al. 2023) and is the source of every non-track-derived feature used in this study.

ERA5 fields were extracted over a domain of 30°N–5°N, 70°E–102°E at three-hourly resolution for all calendar days on which a tracked storm was active, spanning 551 storm-months and approximately 10.5 GB of source NetCDF, and joined to the quality-controlled track rows.

b. Quality control and sample counts

We retain North Indian Ocean storms with genesis dated 1940–2024, applying quality-control criteria that remove records with irregular time steps, coordinate inconsistencies, and track merge or spur artefacts. This yields 852 storms and 27,728 three-hourly track points, the large majority originating in the Bay of Bengal proper with the remainder from Andaman and Arabian Sea genesis.

Multi-horizon target construction, which requires a verified position 12, 24, and 48 h ahead of each origin, imposes a minimum sustained track length. This length requirement, and not a second quality-control pass, excludes 269 short-lived storms and leaves a modelling pool of 583 storms and 15,605 multi-horizon samples. Because train, validation, and test assignment is determined once at the multi-horizon level, all four horizons including 3 h draw from this same storm pool, which keeps the comparison across horizons internally consistent.

c. Storm-wise partitioning

All partitioning is by unique storm identifier rather than by individual observation. This matters because successive positions within one trajectory are strongly autocorrelated, so an observation-level split would place near-duplicate rows on both sides of the partition boundary and inflate apparent skill. For each seed, storms are drawn 70:15:15 into train, validation, and test, giving 408, 87, and 88 storms respectively. Sample counts vary across seeds with the length distribution of the storms drawn, as Table 1 shows.

Seed	Storms			MH samples			3 h samples		
	Train	Val	Test	Train	Val	Test	Train	Val	Test
0	408	87	88	11,066	2,169	2,370	17,594	3,561	3,778
1	408	87	88	11,067	2,425	2,113	17,595	3,817	3,521
2	408	87	88	11,163	2,223	2,219	17,691	3,615	3,627
7	408	87	88	10,352	2,679	2,574	16,880	4,071	3,982
13	408	87	88	11,120	2,262	2,223	17,648	3,654	3,631
99	408	87	88	11,269	1,930	2,406	17,797	3,322	3,814
123	408	87	88	10,693	2,387	2,525	17,221	3,779	3,933
2024	408	87	88	10,730	2,313	2,562	17,258	3,705	3,970
2026	408	87	88	11,081	2,074	2,450	17,609	3,466	3,858

TABLE 1. Held-out split composition. Storm counts are fixed by the 70:15:15 storm-wise draw; sample counts vary with the track-length distribution of the storms assigned. “MH” denotes multi-horizon samples supporting 12, 24, and 48 h targets.

For significance testing we pool held-out test predictions across all nine held-out seeds. The figure of 442 quoted throughout is the number of *distinct* storm identifiers appearing in that pooled test sample; because a storm may fall in the test partition under more than one seed, this is smaller than the sum of the per-seed test counts. Pooled median errors use all pooled test points, while paired per-storm medians are defined on these 442 storms.

d. Feature engineering

Track and physics features (49), used by all variants. Storm position and lagged positions up to three steps back; motion kinematics comprising displacement, translation speed, bearing with sinusoidal encoding, acceleration, and velocity trend; derived physical proxies comprising bearing curvature and stability, seasonal encoding, and a latitude-and-bearing recurvature proxy; native

165 environmental fields comprising distance to land, landfall state, and wind estimate; and indicator
166 flags marking imputed values. This construction follows established practice for kinematic track
167 modelling (Krishnamurti et al. 2010; Lian et al. 2020).

168 **ERA5 point features (14), added in the augmented configuration.** Zonal and meridional wind
169 at 850, 500, and 200 hPa; mean sea level pressure; sea surface temperature; a derived deep-layer
170 steering wind in both components; and derived vertical wind shear as the 200 hPa minus 850 hPa
171 difference in both components together with its magnitude.

172 Sea surface temperature requires specific comment. ERA5 leaves it undefined over land, so
173 near-coast track points have no valid value. We fill these from the nearest valid ocean cell and
174 retain a binary flag marking the substitution. Approximately 38% of rows required this correction,
175 and the affected rows concentrate at higher latitude, more easterly, near-coast positions, which
176 is the signature of a land-masking artefact rather than a bounding-box or data-collection defect.
177 Retaining the flag as a model input lets the learner condition on the substitution rather than treating
178 imputed and observed values identically.

179 A supplementary spatial-patch feature set was also extracted, adding six features per patch size
180 that capture wind-field shear gradient, asymmetry, and nearest-trough proximity within 5° and 3°
181 boxes centred on the storm. It was evaluated but not adopted, for the reasons given in Section a.

182 Because the subset-expert design assumes these engineered features carve the data into physically
183 meaningful regimes rather than arbitrary numerical groupings, we verify that assumption before
184 modelling. Fig. 1 projects the 63-dimensional space onto two dimensions with t-SNE (van der
185 Maaten and Hinton 2008) on a 10,000-sample subset, coloured independently by storm nature,
186 translation speed, latitude band, and displacement magnitude. Coherent structure under all four
187 colourings, rather than under only one, indicates that the feature space separates distinct kinematic
188 and environmental regimes, which is the premise the subset experts of Section 5 exploit.

197 *e. Genesis-window selection*

198 The choice of genesis window trades training-set size against observational quality. We evaluated
199 three candidates, 1940–2024, 1979–2024, and 1990–2024, on nine development seeds using a
200 simple LightGBM baseline. Because both IBTrACS and ERA5 quality improve substantially once
201 satellite observation begins, and because pre-1979 records show near-total missingness in the wind

t-SNE Projection of 63-Dimensional Training Feature Space
($n = 10,000$ training samples | $KL = 1.509$ | perplexity = 40)

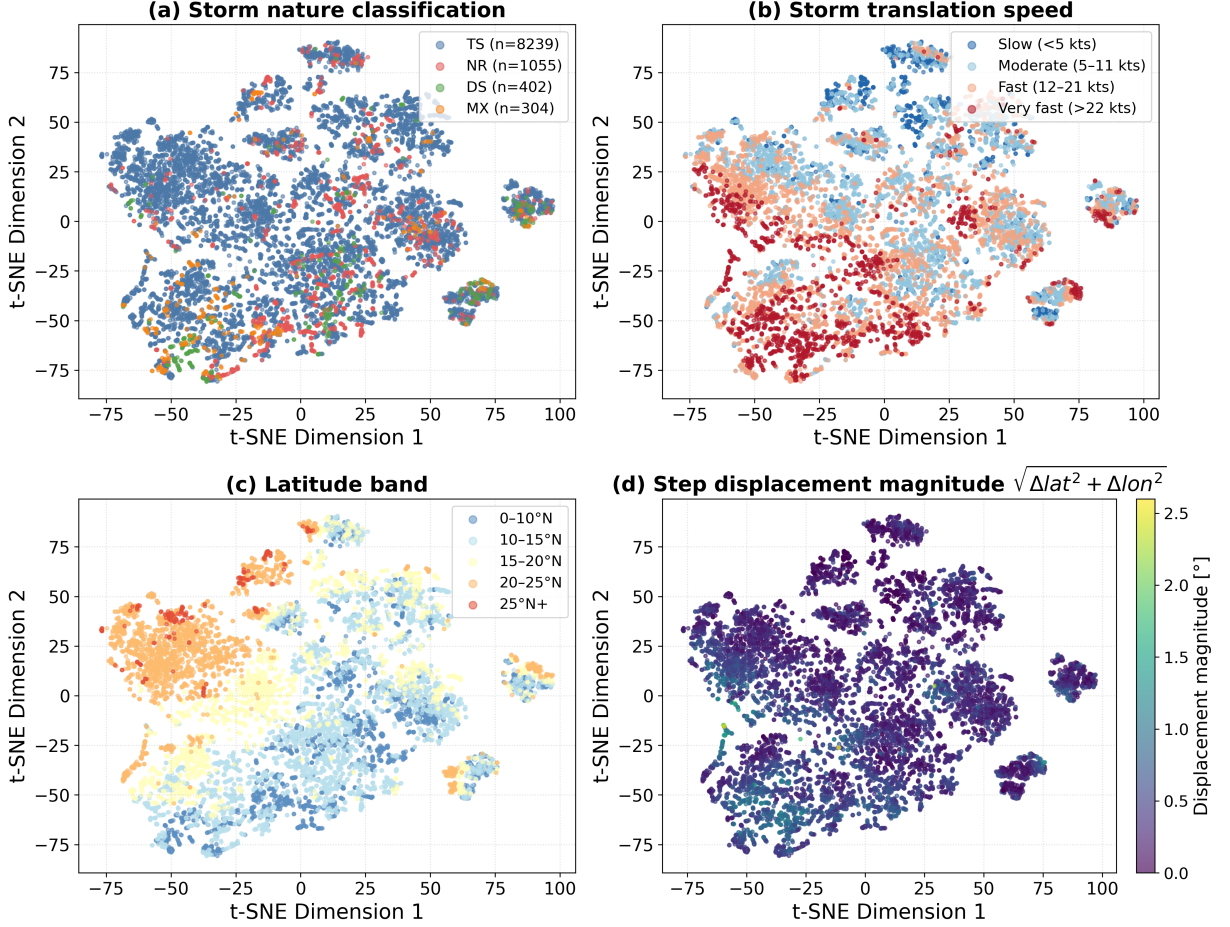


FIG. 1. t-SNE projection (van der Maaten and Hinton 2008) of the 63-dimensional feature space ($n = 10,000$ training samples, held-out seed 0). Colouring by independent meteorological attributes. Coherent separation indicates that the engineered features encode physically distinct regimes, which is the premise of the subset-expert design.

observations used for intensity estimation, a naive comparison risks confounding “more training data” with “easier test set”. We therefore added a fixed-test control in which models trained on each candidate window are all evaluated on one common modern holdout drawn from 1990–2024 genesis, with the same test storms excluded from every training set.

Table 2 gives both comparisons. The wide window wins at every horizon under the fixed-test control, so its advantage is not merely a test-composition artefact. The size of that advantage is

nonetheless strongly horizon-dependent: at 3 h the raw gap of 0.90 km falls to 0.18 km once test composition is held fixed, whereas at 24 and 48 h gaps of 7.9 and 17.7 km persist. We attribute this to storm *position* records remaining comparatively reliable across eras even where wind and intensity observations are sparse, which is the relevant consideration because track geometry, not intensity, is the target variable here. We accordingly lock 1940–2024 for all subsequent modelling.

Training window	3 h	12 h	24 h	48 h
<i>Each window on its own test split</i>				
1940–2024	5.24	36.19	99.77	244.92
1979–2024	6.13	44.04	112.38	271.40
1990–2024	6.14	44.11	108.12	261.58
<i>Fixed common modern holdout</i>				
1940–2024	5.96	39.42	100.18	243.86
1979–2024	6.07	42.24	103.86	249.82
1990–2024	6.14	44.11	108.12	261.58

TABLE 2. Genesis-window comparison, mean test median error in km over nine development seeds with a simple LightGBM baseline. The upper block lets each window use its own test split; the lower block trains on each window and evaluates all three models on one common 1990–2024 genesis holdout.

4. On Test-Set Leakage: A Methodological Note

During initial development, successive architecture revisions, which we refer to internally as Phases 1 through 3, were evaluated on a single fixed train/validation/test split. The decision to advance from one phase to the next was taken by inspecting per-horizon performance gaps on that test set and introducing targeted fusion-path additions intended to close them. No individual test observation was used to fit any model parameter. Nevertheless this iterative, human-guided design loop constitutes test-set leakage: the architecture itself was shaped by repeated exposure to test-set feedback, which invalidates the resulting held-out estimate as a measure of generalisation (Cawley and Talbot 2010).

We detected the pattern by auditing, for each architectural decision point, which data partition had informed it. Within-run decisions, namely fusion-path selection and non-negative least squares weight fitting, proved to be properly restricted to the validation partition. The *cross-run* decision to advance from one phase to the next, however, had been made by inspecting test-set rank tables.

232 The empirical symptom was unambiguous: an architecture that appeared to win at every horizon
233 on the original split did not reproduce that sweep on independent fresh splits.

234 **Correction.** We adopted a strict two-tier seed protocol. Nine *development* seeds,
235 {3,4,5,6,8,9,10,11,14}, carry every architectural and fusion-path decision; during this stage
236 the corresponding held-out partition is neither generated nor examined. Nine *held-out* seeds,
237 {0,1,2,7,13,99,123,2024,2026}, were generated once and evaluated exactly once, after the ar-
238 chitecture was frozen. Architectural convergence is verified by confirming that development-seed
239 and held-out-seed results agree, reported in Section b, rather than by any further tuning against
240 held-out performance.

241 We report this process transparently for two reasons. Iterative test-set inspection is, in our
242 reading, a common and under-scrutinised risk in ensemble-based forecasting benchmarks. And
243 our own uncorrected results, which appeared to show the proposed ensemble winning at every
244 horizon, failed to replicate, which makes the risk concrete rather than hypothetical.

245 5. Methodology

246 a. Baseline systems

247 Seven baselines are benchmarked alongside the proposed ensemble. The **Stacking Ensemble**
248 combines Random Forest, XGBoost, and LightGBM base learners through a Ridge-penalised,
249 cross-validated linear meta-learner. The **Persistence Residual Cascade (PRC)** corrects a naive
250 persistence extrapolation with a dedicated LightGBM residual model fitted in kilometre space
251 and mapped back to degrees. **Random Forest** (Breiman 2001), **LightGBM** (Ke et al. 2017),
252 **XGBoost** (Chen and Guestrin 2016), and **CatBoost** (Prokhorenkova et al. 2018) are included as
253 individual learners. **CB+MotionNN** corrects a CatBoost predictor with a 64-32-16 feedforward
254 network operating on kinematic residual features.

255 b. Proposed architecture: SECE Phase 3

256 For each forecast horizon, SECE constructs four physics-informed feature subsets. At 3 h these
257 are position, motion, environ, and the full feature set; at 12, 24, and 48 h they are motion, physics,
258 environ, and full. Each subset is modelled independently by four tree algorithms, namely CatBoost,
259 XGBoost, LightGBM, and Random Forest, giving sixteen subset-level models per horizon, which

are fused by non-negative least squares into a single subset-consensus signal with separate weights for the latitude and longitude components.

That subset-consensus signal is then combined with the seven baseline predictions, the “champion” pool, in a second-stage non-negative least squares fusion. A horizon-specific set of fusion-path candidates is constructed, listed in Table 3, and a hard validation-best router selects, per seed, whichever candidate achieves the lowest validation median error. The selected candidate’s predictions are what we report. Both the router selection and all weight fitting are restricted to the validation partition, in keeping with Section 4. Fig. 2 summarises the pipeline.

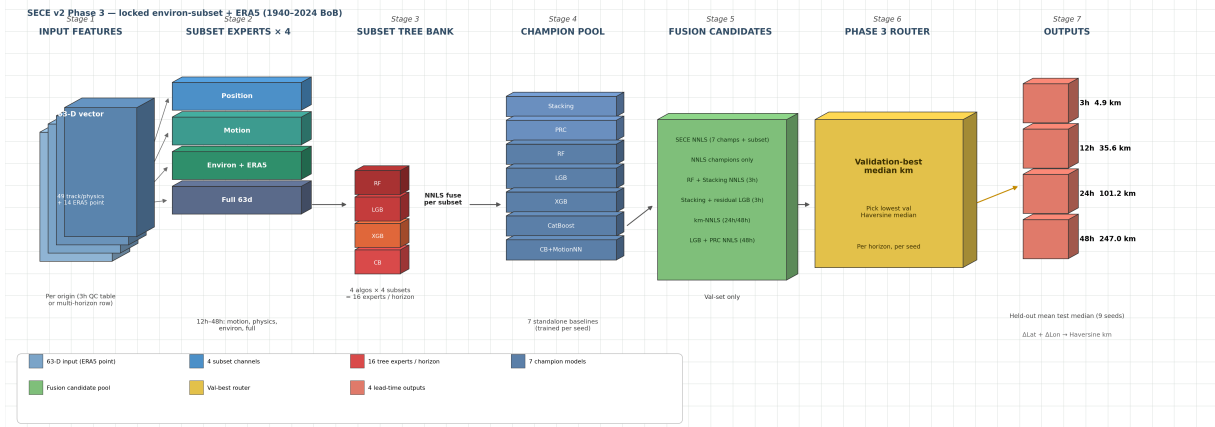


FIG. 2. Locked SECE v2 Phase 3 pipeline. Input is a 63-dimensional vector (49 track/physics features plus 14 point ERA5 fields). Four subset experts, each an NNLS fusion of four tree algorithms, enter a champion pool with seven standalone baselines; a validation-only router selects the fusion candidate with the lowest validation median error at each horizon.

Horizon	Candidate fusion paths
3 h	SECE NNLS full pool; NNLS champions only; RF+Stacking NNLS; Stacking with LightGBM residual correction
12 h	SECE NNLS full pool; NNLS champions only
24 h	SECE NNLS full pool; NNLS champions only; kilometre-space NNLS over champions, persistence-anchored
48 h	as 24 h, plus LightGBM+PRC NNLS and LightGBM+PRC kilometre-NNLS

TABLE 3. Horizon-specific fusion candidates available to the validation-only router. The kilometre-space variant fits weights on persistence-anchored residuals scaled to kilometres and then adds the persistence term back.

The environ expert.

Rather than folding the 14 ERA5 features into existing subsets, we allocate them, together with the native environmental fields for distance to land, landfall state, and storm speed and direction, to a dedicated environ subset with its own four-algorithm expert pool. This choice was made over three tested alternatives, namely ERA5 features added directly to existing point-feature subsets and ERA5 spatial-patch features at 5° and 3° , on the basis of the best balanced performance across all four horizons reported in Section a. It also admits a more direct mechanistic reading, since atmospheric information receives explicit ensemble representation rather than competing for splits inside a larger undifferentiated subset.

c. Training protocol

All tree models use the settings in Table 4, applied uniformly across every system so that measured differences reflect architecture rather than tuning budget (Grinsztajn et al. 2022). Early stopping is disabled for the tree learners.

d. Evaluation metric and statistical testing

The primary metric is median great-circle (Haversine) error in kilometres between predicted and verified storm position, computed per test storm and aggregated (Sinnott 1984). The median rather than the mean is used because per-storm error distributions are strongly right-skewed, so a mean would be dominated by a small number of recurving or rapidly accelerating systems.

For the final held-out evaluation we additionally report 95% confidence intervals on the pooled median obtained from 10,000 storm-level bootstrap resamples (Efron 1979), and paired Wilcoxon signed-rank tests comparing SECE against each of the seven baselines on matched per-storm median errors (Wilcoxon 1945). Resampling and pairing are performed at storm level, not sample level, so that the unit of statistical independence matches the unit of the split. Bonferroni correction is applied across the seven simultaneous comparisons (Bonferroni 1936), giving an adjusted threshold of $\alpha = 0.05/7 \approx 0.0071$.

e. Systems excluded from the held-out table

Before the ERA5-augmented design was frozen we trained CNN-GRU and bidirectional LSTM (Lian et al. 2020; Graves and Schmidhuber 2005) alongside the tree learners on storm-

Component	Setting
Locked architecture	SECE v2 Phase 3 + point ERA5 + environ expert
Genesis window	1940–2024 IBTrACS NI + ERA5 point features
Held-out seeds	0, 1, 2, 7, 13, 99, 123, 2024, 2026
Development seeds (architecture lock only)	3, 4, 5, 6, 8, 9, 10, 11, 14
Split	storm-wise 70:15:15, seed-specific
3 h SECE subsets	position, motion, environ, full
12/24/48 h SECE subsets	motion, physics, environ, full
Subset tree algorithms	CatBoost, XGBoost, LightGBM, Random Forest $\times$ each subset
Subset experts	4 subsets $\times$ 4 algorithms = 16 trees, NNLS-fused to 1 subset signal
Champion / standalone bases	Stacking, PRC, RF, LightGBM, XGBoost, CatBoost, CB+MotionNN
NNLS full pool	7 champions + 1 subset-NNLS signal (degree space, separate lat/lon weights)
Router (Phase 3)	hard val-best: lowest validation median km
3 h extra fusion candidates	SECE NNLS full, NNLS champions, RF+STK NNLS, STK+residual LGB
24 h extra fusion candidates	SECE NNLS full, NNLS champions, km-NNLS (champions, persistence-anchored)
48 h extra fusion candidates	as 24 h plus LGB+PRC NNLS and LGB+PRC km-NNLS
Global learning rate	0.01
Tree $n_{\text{estimators}}$ (RF/XGB/LGB/CB)	300
Tree max depth / CatBoost depth	6
LightGBM <code>num_leaves</code>	63
XGBoost subsample / colsample by tree	0.8 / 0.8
Stacking STK_N (RF+XGB+LGB Ridge)	150
MotionNN	MLP 64-32-16, ReLU, lr 0.01, max_iter 100, early stopping
PRC	persistence displacement + LightGBM residual in km, mapped back to degrees
km-NNLS	NNLS on (pred–persist) $\times$ (111,111) km, then add persist
ERA5 environ keys	u, v at 850/500/200 hPa, MSL, SST, steer u, v , shear u, v , shear magnitude, SST missing flag
Environ subset also includes	DIST2LAND, LANDFALL, NEWDELHI, STORM_SPEED, STORM_DIR + ERA5 keys
Not evaluated on held-out	CNN-GRU, BLSTM, other deep models

TABLE 4. Locked architecture and hyperparameters from `02_architecture_hyperparameters.csv`, applied uniformly across systems unless noted. CNN-GRU and BLSTM were not re-trained on this held-out protocol.

wise development splits, using the same fixed budget philosophy as the locked tree protocol (300 estimators or epochs, learning rate 0.01, no architecture-specific hyperparameter search). In those exploratory runs—track-only features first, then informal checks after ERA5 was added—both recurrent models repeatedly ranked below Random Forest, stacking, and gradient boosting at 3, 12, and 24 h, with short-lead errors closer to persistence-scale displacement than the sub-6 km

318 medians the trees achieved. That pattern is consistent with Grinsztajn et al. (2022) on tabular cy-
319 clone features at basin sample sizes. We therefore scoped the leakage-corrected held-out confirm
320 to the eight tree and hybrid systems already in the production pipeline (Table 6) rather than porting
321 sequence training to nine fresh held-out seeds and a 48 h head. CNN-GRU and BLSTM are absent
322 from every held-out result and figure reported here.

323 6. Results

324 a. *Development-phase ablation*

325 Table 5 reports, for each feature configuration, the number of development seeds out of nine on
326 which SECE achieved the lowest median error among all eight systems, together with its mean
327 median error across those seeds.

328 Two findings motivate the locked architecture. First, ERA5 point features substantially reduce
329 SECE’s mean error at 48 h, from 250.26 to 246.80 km, with 8 of 9 development seeds improving.
330 This establishes that atmospheric information carries genuine, exploitable signal at long lead times
331 rather than merely adding dimensionality. Second, allocating those features to a dedicated environ
332 expert instead of embedding them in existing subsets improves 24 h performance specifically,
333 raising rank-1 frequency from 2/9 to 3/9 and reducing mean error from 99.86 to 99.43 km; 24 h
334 was precisely the horizon at which the plain point-ERA5 configuration was weakest.

335 The same table records a cost that we report rather than suppress. At 3 h the track-only configu-
336 ration retains the lowest mean error, 4.84 against 4.93 km, so ERA5 does not help uniformly across
337 horizons. The short-range gain from ERA5 in this architecture is expressed as rank stability rather
338 than as a lower mean.

339 Spatial-patch features improved short-range rank-1 frequency, reaching 7/9 at both 3 and 12 h, but
340 showed no consistent advantage over the environ-expert configuration at 24 or 48 h, where means
341 fall within roughly 0.4 km of each other, inside the seed-to-seed noise floor (Fig. 3). We therefore
342 did not adopt spatial-patch features into the locked architecture, and report this as a negative result.

361 Fig. 3 shows the same mean-median series as a trend across configurations so that the horizon at
362 which each intervention helps, or fails to help, is visible at a glance.

Configuration	3 h	12 h	24 h	48 h
Track-only (no ERA5)	5/9, 4.84	3/9, 35.48	3/9, 100.48	0/9, 250.26
+ ERA5 point features	5/9, 4.93	7/9, 35.59	2/9, 99.86	1/9, 246.80
+ environ expert (locked)	5/9, 4.93	6/9, 35.56	3/9, 99.43	2/9, 246.94
+ spatial patch (5°)	7/9, 4.93	7/9, 35.56	1/9, 99.74	1/9, 246.55
+ spatial patch (3°)	7/9, 4.93	7/9, 35.66	2/9, 99.42	3/9, 247.00
Held-out confirm (locked)	6/9, 4.90	6/9, 35.63	2/9, 101.21	1/9, 246.99

TABLE 5. Development-seed ablation over nine development seeds. Each cell gives the SECE rank-1 count out of nine, then the mean median error in km. The locked configuration is in bold. The final row is the independent held-out confirm on disjoint seeds, included here only to show tier agreement.

b. Held-out evaluation

Table 6 reports pooled median errors and bootstrap confidence intervals for all eight systems at all four horizons, over 34,114 test samples at 3 h and 21,442 at the longer horizons.

At 3 h SECE leads with 4.895 km, ahead of Random Forest at 4.927 km and the Stacking Ensemble at 4.945 km, with a clear gap to the remaining five systems. At 12 h SECE again leads with 35.538 km against 36.146 km for stacking. At 24 h the ordering changes: stacking and XGBoost take the top two point estimates, 100.771 and 101.029 km, with SECE third at 101.186 km. At 48 h XGBoost and PRC lead with 242.702 and 243.229 km against 246.331 km for SECE. In both long-lead cases the confidence intervals overlap almost entirely, which is why the paired tests rather than the point estimates carry the inferential weight.

Table 7 gives those tests in full, with Bonferroni-adjusted p -values and the rank-biserial correlation as an effect size. At 3 and 12 h all seven comparisons are significant after correction, so SECE beats every baseline at both short horizons. At 24 h only CatBoost and CB+MotionNN remain significant; Random Forest is significant raw but not after correction, and stacking, PRC, LightGBM, and XGBoost are not significant by either standard. At 48 h only CatBoost remains significant, with stacking and CB+MotionNN significant raw but not after correction.

The effect-size column carries an important detail. Every rank-biserial correlation in the table is negative, which under our sign convention means SECE holds the lower per-storm error in the paired comparison. Consequently SECE is never significantly *worse* than any baseline at any horizon; at long lead the correct statement is parity, not defeat. The magnitudes also make the

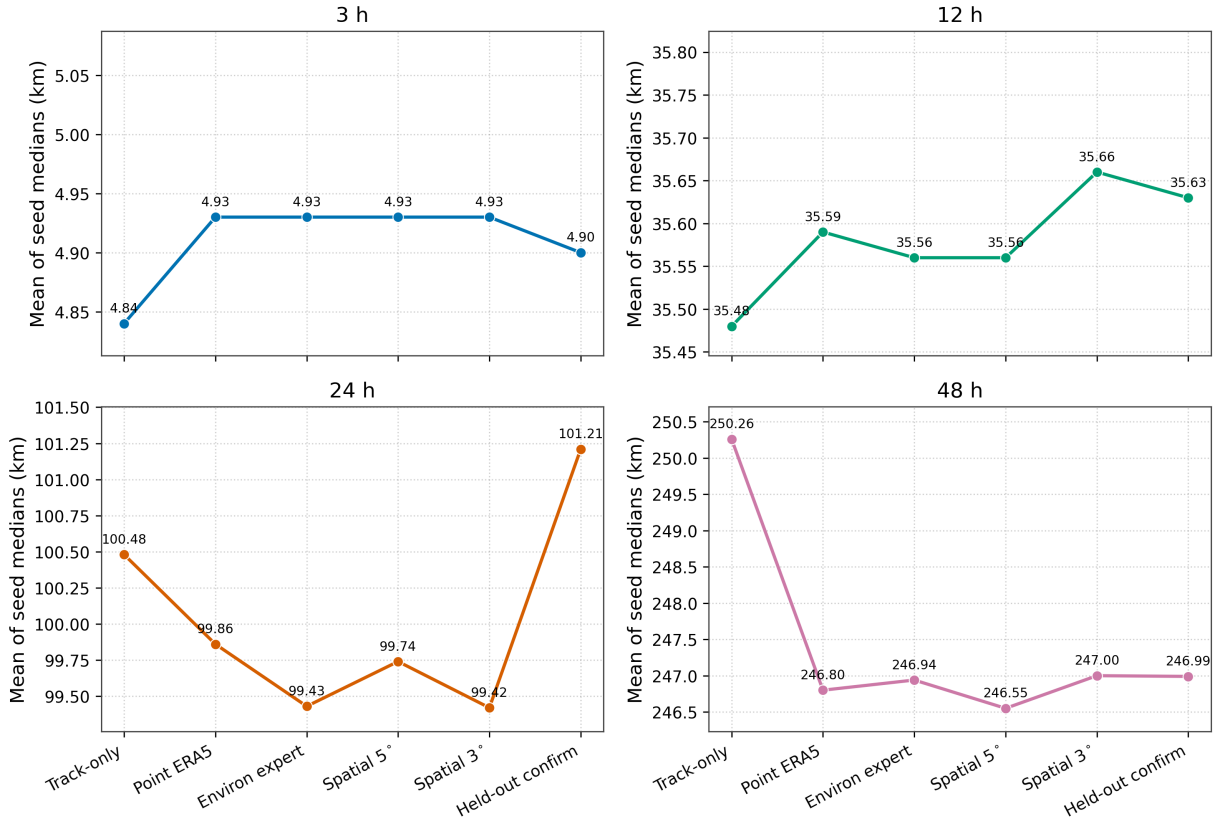


FIG. 3. Development ablation trend from `full_experiment_summary_table.csv`. The y-axis is the mean of SECE test medians across nine seeds (km). Configurations on the x-axis are sequential interventions on the development seeds except the last point, which is the independent held-out confirm on nine disjoint seeds (not a sixth architecture). Each panel uses its own vertical scale so that 3 h changes of a few tenths of a kilometre remain visible. Point ERA5 helps at 48 h; the environ expert is the locked 24 h compromise; spatial 5°/3° patches do not justify adoption over the environ expert.

horizon dependence visible: against stacking the effect size falls from -0.208 at 3 h to -0.071 at 24 h, and against XGBoost from -0.993 to -0.003 at 48 h.

Pooled medians summarise central tendency but hide how often each system wins outright. Table 8 therefore counts, per horizon, how many of the nine held-out seeds each system led. The pattern reinforces the significance tests: SECE takes 6 of 9 seeds at both 3 and 12 h, but only 2 of 9 at 24 h and 1 of 9 at 48 h, where PRC and XGBoost lead more often. A system can hold the best pooled median while winning a minority of seeds, and both views are needed to describe long-lead behaviour honestly.

Horizon	System	Median (km)	95% CI
3 h	SECE v2 Phase 3	4.895	[4.714, 5.079]
	Random Forest	4.927	[4.734, 5.125]
	Stacking Ensemble	4.945	[4.763, 5.123]
	CB+MotionNN	5.813	[5.701, 5.928]
	LightGBM	6.026	[5.889, 6.168]
	XGBoost	6.411	[6.243, 6.580]
	CatBoost	8.479	[8.284, 8.691]
	PRC	8.872	[8.707, 9.058]
12 h	SECE v2 Phase 3	35.538	[34.208, 36.925]
	Stacking Ensemble	36.146	[34.744, 37.701]
	Random Forest	36.901	[35.525, 38.323]
	LightGBM	36.928	[35.498, 38.338]
	XGBoost	36.947	[35.531, 38.416]
	PRC	37.177	[35.704, 38.527]
	CB+MotionNN	37.413	[36.047, 38.949]
	CatBoost	42.028	[40.381, 43.656]
24 h	Stacking Ensemble	100.771	[96.887, 104.703]
	XGBoost	101.029	[96.932, 105.286]
	SECE v2 Phase 3	101.186	[97.301, 104.808]
	LightGBM	101.632	[97.778, 105.589]
	PRC	101.941	[98.131, 105.824]
	Random Forest	103.511	[99.652, 107.437]
	CB+MotionNN	104.034	[100.239, 108.074]
	CatBoost	107.191	[102.555, 111.839]
48 h	XGBoost	242.702	[232.596, 255.403]
	PRC	243.229	[231.650, 255.158]
	LightGBM	245.975	[234.926, 257.214]
	Stacking Ensemble	245.977	[235.938, 257.183]
	SECE v2 Phase 3	246.331	[235.541, 257.614]
	CB+MotionNN	249.678	[238.972, 261.153]
	Random Forest	251.305	[240.752, 262.324]
	CatBoost	254.862	[242.991, 267.797]

TABLE 6. Held-out test performance: pooled median great-circle error in km with 95% bootstrap confidence interval from 10,000 storm-level resamples. Nine held-out seeds pooled; 442 distinct storm identifiers. Eight tree/hybrid systems only; CNN-GRU and BLSTM omitted (section e).

Fig. 4 places the eight systems in the mean-error versus accuracy-index plane at all four horizons (mean on the horizontal axis; medians in Table 6). At 3 and 12 h the leading systems separate visibly from the remainder; at 24 and 48 h they collapse onto a common cluster.

Baseline	3 h		12 h		24 h		48 h	
	p	r	p	r	p	r	p	r
Stacking Ensemble	0.0016	-0.208	0.0014	-0.210	1.000	-0.071	0.262 [†]	-0.114
Random Forest	0.0272	-0.164	1.1×10^{-5}	-0.263	0.0737 [†]	-0.140	0.704	-0.090
LightGBM	$< 10^{-6}$	-0.970	$< 10^{-6}$	-0.306	1.000	-0.044	1.000	-0.068
XGBoost	$< 10^{-6}$	-0.993	$< 10^{-6}$	-0.364	1.000	-0.046	1.000	-0.003
PRC	$< 10^{-6}$	-1.000	$< 10^{-6}$	-0.394	0.898	-0.084	1.000	-0.005
CatBoost	$< 10^{-6}$	-1.000	$< 10^{-6}$	-0.806	$< 10^{-6}$	-0.450	$< 10^{-6}$	-0.352
CB+MotionNN	$< 10^{-6}$	-0.833	$< 10^{-6}$	-0.443	1×10^{-5}	-0.265	0.262 [†]	-0.114

TABLE 7. Paired Wilcoxon signed-rank tests of SECE against each baseline on 442 matched per-storm medians, reporting the Bonferroni-adjusted p -value and the rank-biserial correlation r as an effect size. Bold marks a significant SECE advantage at the adjusted threshold $\alpha \approx 0.0071$; a dagger marks significance before correction only. Every r is negative, meaning SECE holds the lower per-storm error, so no baseline is significantly better than SECE at any horizon. The collapse of $|r|$ from 3 to 48 h is the quantitative signature of the horizon dependence discussed in Section 7.

Horizon	Seeds led (out of nine)
3 h	SECE 6, Random Forest 2, Stacking 1
12 h	SECE 6, Stacking 3
24 h	Stacking 3, SECE 2, LightGBM 2, XGBoost 2
48 h	PRC 4, XGBoost 3, SECE 1, Stacking 1

TABLE 8. Number of held-out seeds, out of nine, on which each system achieved the lowest test median error. Systems not listed for a horizon led on no seed.

c. Per-storm ERA5 contribution at long horizons

To characterise the distribution of ERA5’s benefit independently of the ensemble machinery, we compare a single LightGBM model trained with and without ERA5 features, storm by storm, on a representative development seed with 88 test storms. At 24 h ERA5 improves accuracy for 43 of 88 storms, or 48.9%, with a near-zero mean effect, indicating rough parity at this horizon for this simpler model. At 48 h it improves accuracy for 54 of 88 storms, or 61.4%, with a median per-storm improvement of 10.9 km (Fig. 5).

The benefit therefore strengthens with lead time. This is consistent with the mechanistic expectation that a storm’s own recent kinematic history becomes a progressively weaker predictor of its

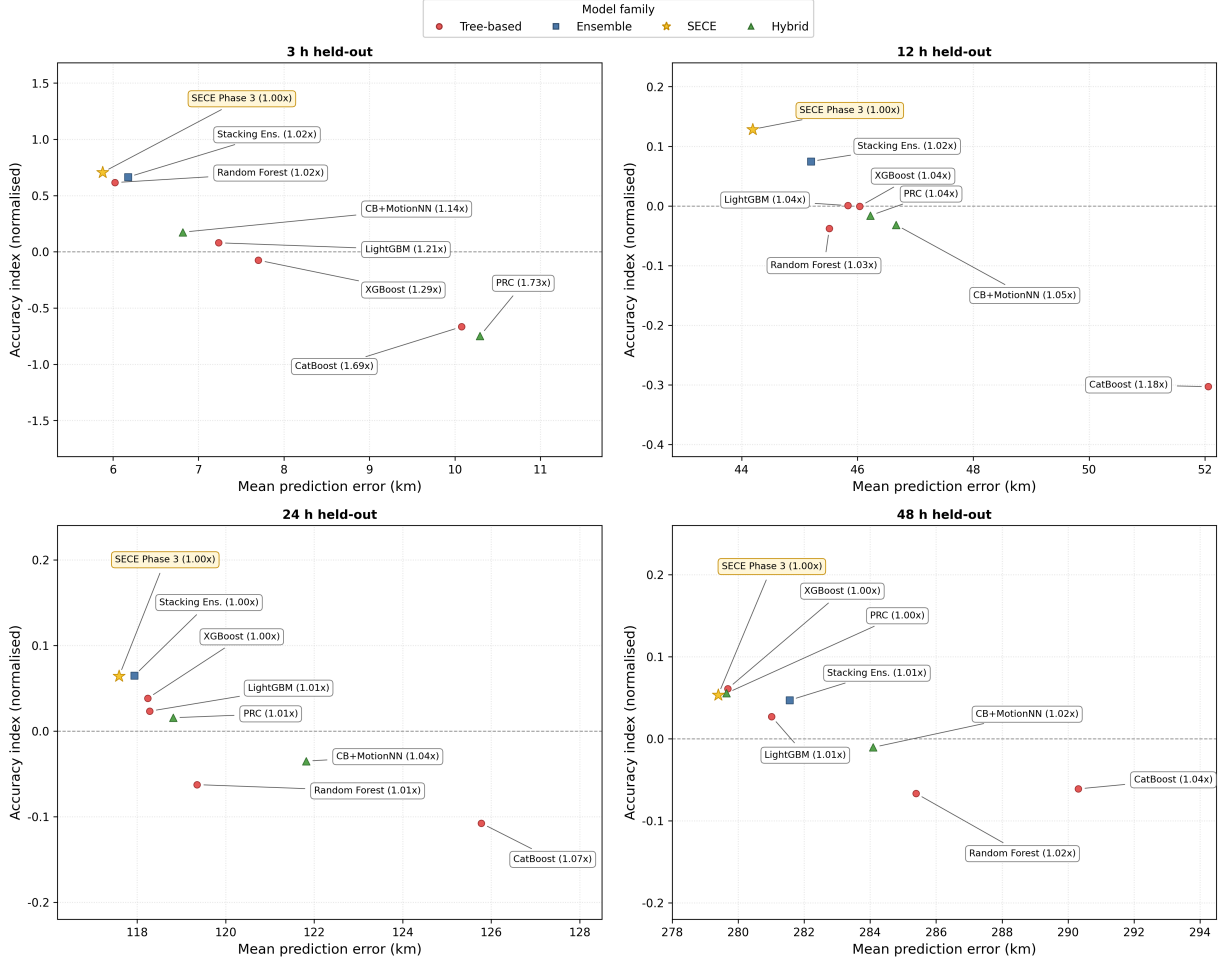


FIG. 4. Held-out four-horizon benchmark for the eight locked systems. Each point is computed from the pooled nine-seed held-out prediction export (results/sece_era5_envirion_heldout/task_a_test_predictions/), audited in benchmark_*h_heldout_stats.json. The horizontal axis is *mean* great-circle error (km) over all scored test origins; parenthetical multipliers are that system's mean divided by the SECE mean. The vertical axis is an accuracy index derived from *median* error, normalised so that the tree-family median sits at zero. Table 6 reports the corresponding medians with bootstrap confidence intervals. Mean and median therefore answer different questions and are not interchangeable.

future position, so large-scale steering information becomes proportionally more valuable (Giffard-Roisin et al. 2020; Goerss 2004).

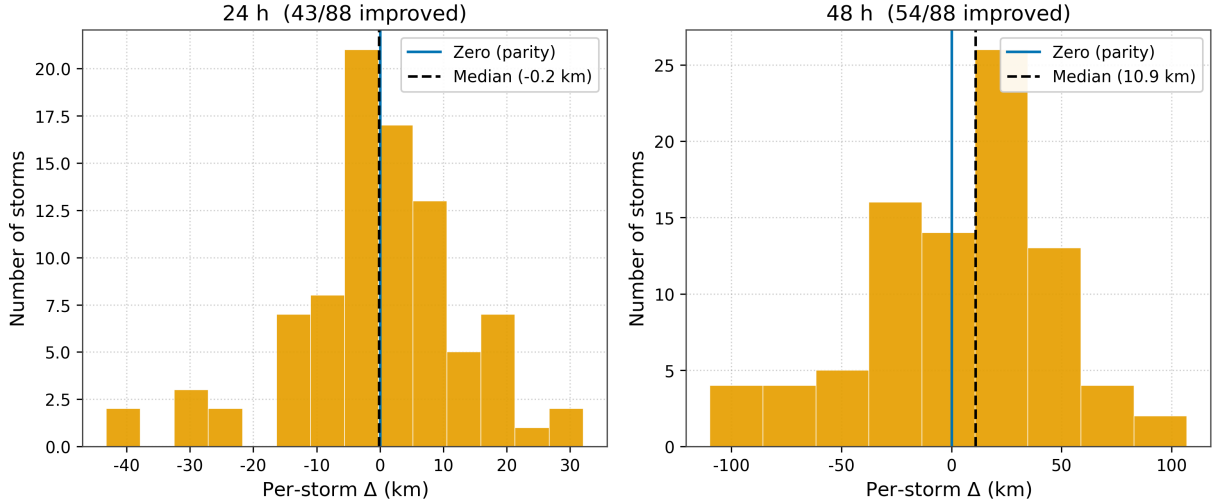


FIG. 5. Per-storm LightGBM diagnostic comparing track-only error with point-ERA5 error on held-out seed 0 (05_seed0_lgb_era5_vs_trackonly_per_storm.csv). Δ is track-only median error minus ERA5-informed median error, so positive values mean ERA5 improved that storm. Blue: zero (parity); black dashed: median Δ . Panel titles report $n_{\text{improved}}/n_{\text{total}}$. This is a single-model, single-seed diagnostic, not the full SECE stack.

d. Feature importance within the environ expert

Table 9 and Fig. 6 report gain-based importance for the environ-subset expert, from a diagnostic retrain on one held-out seed under the locked LightGBM specification, averaged across the latitude and longitude output heads. This is a single-seed tree-gain diagnostic, not a SHAP or multi-seed attribution, and we present it only as an indication of which fields that particular expert relies upon.

Two observations survive that caveat. First, kinematic features retain the dominant share of gain even inside this ERA5-inclusive expert: storm direction and speed are the two highest-ranked features at both horizons, which underscores that atmospheric information supplements rather than displaces kinematic signal. Second, every ERA5 field's share of gain grows markedly from 24 to 48 h, with sea surface temperature rising from 1.66% to 4.15% and 200 hPa zonal wind from 1.34% to 3.32%, which is an independent signature of the same lead-time dependence found in Section c.

We deliberately do not interpret rank order within the shear and steering group. Those components are mutually collinear by construction, since shear is defined as a difference of the same wind levels that enter the steering term, so gain-based importance may apportion credit arbitrarily

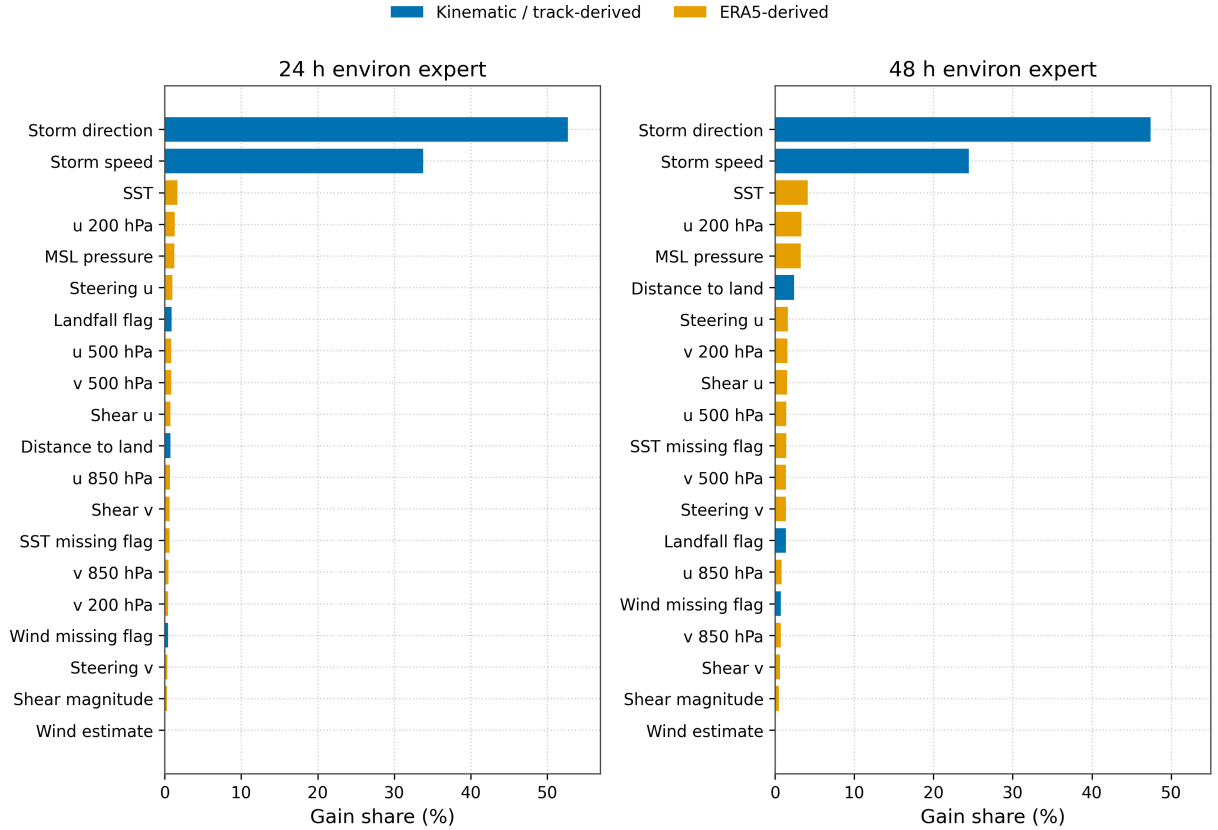


FIG. 6. LightGBM gain share for every feature in the environ-subset expert at 24 and 48 h, from 04_environ_expert_feature_importance.csv. Blue: kinematic / track-derived features; orange: ERA5-derived features, the colour convention used throughout this manuscript. Bars are sorted by gain share. This is held-out seed 0 only, gain-based (not SHAP or permutation importance), and environ-expert-only (not the full SECE stack).

among them. Dropping one wind level or replacing the expert with ridge regression would remove the physical shear–steering decomposition the environ subset is meant to preserve; grouped permutation or SHAP tests would be needed for stable attribution within that block. Our claim is the lead-time *growth* of total ERA5 gain (Section c), not the within-group rank order.

e. Case studies

Fig. 7 shows predicted and observed tracks for a held-out storm. Each marker is a prediction issued from a distinct origin time along the trajectory and offset forward by the panel’s lead time, so the panels show direct multi-horizon endpoints rather than a recursively rolled-out track. Where a

ERA5 field	Share at 24 h	Share at 48 h
Sea surface temperature (sst)	0.0166	0.0415
200 hPa zonal wind (u200)	0.0134	0.0332
Mean sea level pressure (msl)	0.0128	0.0323
Steering wind, zonal (steer_u)	0.0101	0.0161
500 hPa zonal wind (u500)	0.0086	0.0141
500 hPa meridional wind (v500)	0.0085	0.0137
Shear, zonal (shear_u)	0.0078	0.0152
850 hPa zonal wind (u850)	0.0070	0.0084
Shear, meridional (shear_v)	0.0067	0.0064
SST imputation flag (sst_missing)	0.0067	0.0140
850 hPa meridional wind (v850)	0.0048	0.0074
200 hPa meridional wind (v200)	0.0044	0.0156
Steering wind, meridional (steer_v)	0.0036	0.0135
Shear magnitude (shear_mag)	0.0032	0.0046

TABLE 9. Gain-based importance share of ERA5 fields within the environ-subset expert, single-seed diagnostic. Shares are relative to total environ-subset gain, so they do not sum to one; kinematic features, which rank above all ERA5 fields, account for the remainder.

competitor is drawn, it is the system with the lowest *pooled* held-out median at that horizon, namely Random Forest at 3 h, stacking at 12 and 24 h, and XGBoost at 48 h, which is not necessarily the best system on the individual storm shown. The distinction matters in practice, as the following two cases demonstrate.

Cyclone Laila, 2010. Among the 88 test storms on the diagnostic seed, Laila exhibits the largest single-storm ERA5 benefit for the LightGBM diagnostic of Section c: a 25.0 km improvement at 24 h and 107.0 km at 48 h when ERA5 features are included. We report this as an illustration of that mechanism, a plain gradient-boosted model’s response to atmospheric information, and not as a claim about SECE’s performance on this storm. Separately, at the ensemble level, SECE’s own per-storm error on Laila is 111.3 km at 24 h, locally better than the pooled-best competitor at 128.9 km, while at 48 h the pattern reverses and XGBoost is locally stronger, 204.9 against 236.1 km, consistent with the pooled 48 h leaderboard.

A representative 1996 system. To illustrate typical rather than extreme behaviour we select the test storm whose SECE per-storm median error is closest to that seed’s overall SECE 24 h median of 106.43 km. This is an unnamed 1996 storm tracking through the central Bay of Bengal with genesis near 8.4°N, 87.7°E, whose per-storm median error of 107.24 km closely matches

Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

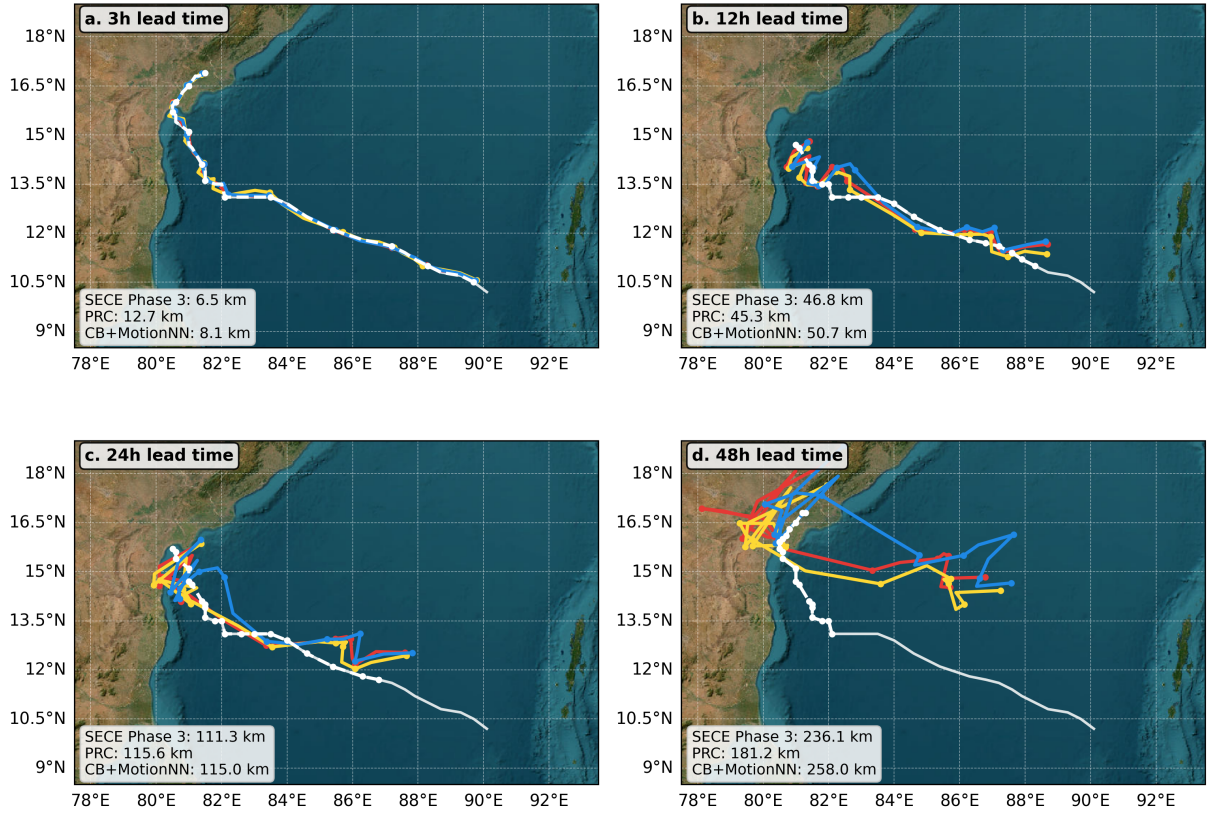


FIG. 7. Cyclone Laila (2010) on held-out seed 0 (05_case_study_trajectories.csv). Four panels: 3, 12, 24, and 48 h lead times. White solid line: storm centre at each forecast origin; white dashed with markers: verified IBTrACS position at that lead. Red, yellow, and blue tracks are SECE Phase 3, PRC, and CB+MotionNN endpoints from each origin (not a recursively rolled-out trajectory). Panel legends give median great-circle error (km) for those three systems on this storm.

the typical-case benchmark. On this storm SECE at 107.2 km and the pooled-best competitor at 110.7 km are closely matched at 24 h, while XGBoost shows a wider local advantage at 48 h, 276.0 against 306.3 km. Both cases are consistent with the aggregate finding that SECE's advantage concentrates at shorter horizons.

7. Discussion

The central empirical finding of this study is horizon-dependent. SECE’s context-aware, ensemble-of-experts architecture provides a statistically robust advantage at short to medium lead times, 3 and 12 h, where storm-level context such as recurvature proxies, coastal proximity, and atmospheric steering carries strong learnable signal. That advantage does not persist as statistically distinguishable superiority at 24 and 48 h, where SECE instead achieves parity with, and never significant inferiority to, the strongest available baselines.

We interpret this pattern from a bias-variance perspective on ensemble complexity. As target uncertainty grows with lead time, the marginal value of additional ensemble sophistication diminishes, and a comparatively simpler, well-regularised stacking approach becomes competitive. This extends the finding of Grinsztajn et al. (2022), that tree-based methods outperform deep learning on tabular data at this scale, by one level: *within* the tree-based ensemble family, complexity itself should be calibrated to the information content available at a given forecast horizon rather than applied uniformly (Ganaie et al. 2022). The effect sizes in Table 7 trace that calibration quantitatively, collapsing from -0.208 against stacking at 3 h to -0.071 at 24 h.

ERA5 features provide genuine value, but the locus of that value differs from where SECE’s architectural advantage is clearest. The per-storm LightGBM diagnostic of Section c places ERA5’s majority-of-storms benefit at 48 h, and the gain shares of Section d agree, roughly tripling from 24 to 48 h. The environ-expert architectural addition, by contrast, delivered its clearest ensemble-level gain at 24 h. We read this as genuine if imperfectly aligned confirmation that atmospheric information and ensemble architecture contribute complementary rather than identical improvements, and we flag the alignment between them as a direction for further investigation rather than a discrepancy to be smoothed over.

For operational interpretation, the horizons at which SECE is significantly better are also the horizons that govern short-fuse decisions such as port closure and shelter activation, whereas 24 to 48 h guidance is better treated as a consensus problem in which several strong systems are statistically interchangeable (Goerss 2000, 2004). A practitioner choosing a single system for this basin could reasonably run SECE for short-range guidance and a simpler stack or persistence-residual model at long range, and our tables give the evidence for that split rather than a blanket recommendation.

Operational context clarifies what those medians do and do not imply. RSMC New Delhi verifies official tropical-cyclone track forecasts over the North Indian Ocean using WMO-standard direct position error against IMD best track (Mohapatra et al. 2013). For verified forecasts in 2009–2013, published mean errors were 124.1 km at 24 h and 202.1 km at 48 h (Mohapatra et al. 2013). Our held-out SECE pooled medians are 101.2 and 246.3 km at the same leads. Those figures are *not* directly comparable: we score retrospective IBTrACS origins with a fixed ML hindcast, whereas IMD errors average operational issuances from deep depression onward on a year-specific storm sample. We therefore do not claim to beat the operational multi-model ensemble or dynamical–statistical Cyclone Prediction System; this confirm benchmarks learned systems against each other. The comparison shows only that 24 h ML errors sit in the same order of magnitude as published official 24 h verification, while 48 h ML medians remain broadly consistent with the difficulty of extended-range track guidance in this basin.

Finally, the leakage correction of Section 4 is not an ancillary methodological remark but the reason the preceding claims are credible. Our pre-correction architecture appeared to win at every horizon, and that apparent sweep did not replicate on fresh seeds. The agreement between the development and held-out tiers after freezing, 246.94 against 246.99 km at 48 h, is the evidence that the locked design generalises, and it is evidence of a kind that only a firewalled confirm can produce (Cawley and Talbot 2010).

8. Limitations

Environmental fields. The locked model uses ERA5 only at the storm centre (14 point variables). We tested 5° and 3° spatial-patch statistics in development; they did not improve 24–48 h mean error enough to justify adoption (Table 5; Fig. 3), remaining inside seed noise relative to the environ expert. Gridded or attention-based encoders therefore remain outside this benchmark.

Deterministic outputs. Every system returns a single track point without forecast spread. That omission matters most at 24–48 h, where Table 6 shows overlapping leading medians and right-skewed errors. A probabilistic formulation trained under a proper scoring rule (Gneiting and Raftery 2007) is future work, not a result claimed here.

Baseline scope. The confirm table is an eight-way tree and hybrid comparison. CNN-GRU and BLSTM were not re-trained on the locked ERA5 protocol; section e summarises exploratory

579 development under the uniform budget (Grinsztajn et al. 2022). Tuned deep encoders on gridded
580 ERA5 remain future work.

581 **Historical mixing.** Genesis 1940–2024 spans pre-satellite and modern eras. A fixed modern
582 holdout (section e) supports the wide window for track error, yet pre-1979 wind-observation
583 sparsity limits intensity-adjacent interpretation, and residual confounding between sample size and
584 observational era cannot be fully excluded.

585 **Basin scope and transfer.** Feature subsets, router menus, and confirm statistics are tuned to
586 North Indian Ocean genesis and Bay of Bengal steering regimes (monsoon trough interaction,
587 shallow funnel geometry, land proximity). We report no zero-shot or retrained evaluation on the
588 Western North Pacific, Atlantic, or other basins; cross-basin transfer remains strictly future work.

589 **Attribution.** Section d and Fig. 6 report LightGBM *gain* on the environ expert for held-out
590 seed 0 only. This is illustrative, not multi-seed SHAP or permutation testing, and not mechanistic
591 proof of how atmospheric fields cause track-error reductions.

592 9. Conclusion

593 We present a leakage-corrected, statistically validated, ERA5-augmented benchmark for multi-
594 horizon tropical cyclone track prediction over the Bay of Bengal. The proposed ensemble, SECE
595 v2 Phase 3, significantly outperforms seven baseline systems at 3 and 12 h lead times and achieves
596 statistical parity with the strongest baselines at 24 and 48 h, never falling to significant inferiority
597 at any tested horizon among the evaluated tree-based systems. We identify and correct a test-set
598 leakage failure mode encountered during our own development, document a controlled ablation
599 isolating the contribution of ERA5 atmospheric features, and report negative results, specifically
600 the rejection of spatial-patch features, alongside the positive ones. The intent is a complete
601 and reproducible account of what this class of model can and cannot currently achieve for an
602 operationally important and underserved region.

603 *Acknowledgments.* ERA5 reanalysis fields were obtained from the Copernicus Climate Data Store
604 (Hersbach et al. 2023). IBTrACS best-track data were obtained from NOAA’s National Centers
605 for Environmental Information (Knapp et al. 2010; Gahtan et al. 2024). The authors declare no
606 conflicts of interest.

Data availability statement. Processed modelling tables, locked held-out predictions, bootstrap and Wilcoxon outputs, case-study trajectories, seed lists, and split counts are versioned with the project. ERA5 is available from the Copernicus Climate Data Store (<https://doi.org/10.24381/cds.adbb2d47>). IBTrACS version 4r01 is available from NOAA NCEI (<https://doi.org/10.25921/82ty-9e16>). Every new figure in this manuscript regenerates from those frozen files.

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